

React Authentication Flow Application

1. Package.json (Dependencies)

json

```
{
  "name": "react-auth-flow",
  "version": "1.0.0",
  "private": true,
  "dependencies": {
    "react": "^18.2.0",
    "react-dom": "^18.2.0",
    "react-router-dom": "^6.8.0",
    "react-scripts": "5.0.1"
  }
}
```

2. App.js (Main Application)

jsx

```
import React, { createContext, useState, useContext } from 'react';
import { BrowserRouter, Routes, Route, Link, Navigate } from 'react-router-dom';
import './App.css';
import Home from './components/Home';
import Login from './components/Login';
import Dashboard from './components/Dashboard';
```

```
// Create Auth Context
export const AuthContext = createContext();
```

```
// Auth Provider Component
export const AuthProvider = ({ children }) => {
  const [user, setUser] = useState(null);

  const login = (username, password) => {
    // Simulate authentication - in real app, this would call an API
    if (username && password) {
      setUser({ username, email: `${username}@example.com` });
      return true;
    }
    return false;
  };
};
```

```
const logout = () => {
  setUser(null);
};

return (
  <AuthContext.Provider value={{ user, login, logout }}>
    {children}
  </AuthContext.Provider>
);
};
```

```
function App() {
  return (
    <AuthProvider>
      <BrowserRouter>
        <div className="App">
          <nav className="navbar">
            <div className="nav-brand">
              <h2>Auth Flow Demo</h2>
            </div>
            <ul className="nav-menu">
              <li>
                <Link to="/" className="nav-link">Home</Link>
              </li>
              <li>
                <Link to="/dashboard" className="nav-link">Dashboard</Link>
              </li>
            </ul>
          </nav>

          <div className="container">
            <Routes>
```

```

        <Route path="/" element={<Home />} />
        <Route path="/login" element={<Login />} />
        <Route path="/dashboard" element={<Dashboard />} />
      </Routes>
    </div>
  </div>
</BrowserRouter>
</AuthProvider>
);
}

export default App;

```

3. App.css (Styling)

```

CSS

* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
  min-height: 100vh;
}

.App {
  min-height: 100vh;
}

/* Navbar Styles */
.navbar {
  background: white;
  padding: 1rem 2rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
  box-shadow: 0 2px 10px rgba(0,0,0,0.1);
}

.nav-brand h2 {
  color: #667eea;
  font-size: 1.5rem;
}

.nav-menu {
  display: flex;
  list-style: none;
  gap: 2rem;
}

.nav-link {
  text-decoration: none;
  color: #333;
  font-weight: 500;
  padding: 0.5rem 1rem;
  border-radius: 4px;
  transition: all 0.3s ease;
}

.nav-link:hover {
  background: #667eea;
  color: white;
}

/* Container Styles */
.container {
  max-width: 1200px;
  margin: 2rem auto;
  padding: 0 1rem;
}

/* Page Styles */
.page {
  background: white;
  border-radius: 10px;
  padding: 2rem;
}

```

```
    box-shadow: 0 5px 20px rgba(0,0,0,0.1);
}

.page h1 {
    color: #667eea;
    margin-bottom: 1.5rem;
    font-size: 2.5rem;
}

.page p {
    color: #666;
    line-height: 1.6;
    margin-bottom: 1rem;
}

/* Login Form Styles */
.login-container {
    max-width: 400px;
    margin: 0 auto;
}

.login-form {
    background: white;
    padding: 2rem;
    border-radius: 10px;
    box-shadow: 0 5px 20px rgba(0,0,0,0.1);
}

.form-group {
    margin-bottom: 1.5rem;
}

.form-group label {
    display: block;
    margin-bottom: 0.5rem;
    color: #333;
    font-weight: 500;
}

.form-group input {
    width: 100%;
    padding: 0.75rem;
    border: 1px solid #ddd;
    border-radius: 4px;
    font-size: 1rem;
    transition: border-color 0.3s ease;
}

.form-group input:focus {
    outline: none;
    border-color: #667eea;
}

.login-btn {
    width: 100%;
    background: #667eea;
    color: white;
    border: none;
    padding: 1rem;
    font-size: 1rem;
    border-radius: 4px;
    cursor: pointer;
    transition: background 0.3s ease;
    margin-bottom: 1rem;
}

.login-btn:hover {
    background: #764ba2;
}

.demo-credentials {
    text-align: center;
    padding: 1rem;
    background: #f8f9fa;
    border-radius: 4px;
    font-size: 0.9rem;
    color: #666;
}

.demo-credentials p {
    margin: 0.25rem 0;
}
```

```

}

/* Dashboard Styles */
.dashboard-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 2rem;
  padding-bottom: 1rem;
  border-bottom: 2px solid #f0f0f0;
}

.user-info {
  background: #f8f9fa;
  padding: 0.5rem 1rem;
  border-radius: 20px;
  color: #667eea;
  font-weight: 500;
}

.logout-btn {
  background: #dc3545;
  color: white;
  border: none;
  padding: 0.5rem 1.5rem;
  border-radius: 4px;
  cursor: pointer;
  font-size: 1rem;
  transition: background 0.3s ease;
}

.logout-btn:hover {
  background: #c82333;
}

/* Stats Cards */
.stats-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
  gap: 1.5rem;
  margin-bottom: 2rem;
}

.stat-card {
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
  color: white;
  padding: 1.5rem;
  border-radius: 8px;
  text-align: center;
}

.stat-card h3 {
  font-size: 1rem;
  margin-bottom: 0.5rem;
  opacity: 0.9;
}

.stat-card .stat-value {
  font-size: 2rem;
  font-weight: bold;
}

/* Activity List */
.activity-list {
  background: #f8f9fa;
  padding: 1.5rem;
  border-radius: 8px;
}

.activity-list h3 {
  color: #333;
  margin-bottom: 1rem;
}

.activity-list ul {
  list-style: none;
}

.activity-list li {
  padding: 0.75rem;
  border-bottom: 1px solid #ddd;
}

```

```
    color: #666;
}

.activity-list li:last-child {
    border-bottom: none;
}

/* Home Page Styles */
.features-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
    gap: 1.5rem;
    margin: 2rem 0;
}

.feature-card {
    text-align: center;
    padding: 1.5rem;
    background: #f8f9fa;
    border-radius: 8px;
    transition: transform 0.3s ease;
}

.feature-card:hover {
    transform: translateY(-5px);
}

.feature-icon {
    font-size: 2.5rem;
    margin-bottom: 1rem;
}

.feature-card h3 {
    color: #333;
    margin-bottom: 0.5rem;
}

.feature-card p {
    color: #666;
    font-size: 0.9rem;
    margin: 0;
}

.cta-section {
    text-align: center;
    margin-top: 2rem;
    padding: 2rem;
    background: #f8f9fa;
    border-radius: 8px;
}

.cta-btn {
    display: inline-block;
    background: #667eea;
    color: white;
    text-decoration: none;
    padding: 1rem 2rem;
    border-radius: 4px;
    font-weight: 500;
    transition: background 0.3s ease;
}

.cta-btn:hover {
    background: #764ba2;
}

/* Responsive Design */
@media (max-width: 768px) {
    .navbar {
        flex-direction: column;
        padding: 1rem;
    }

    .nav-brand {
        margin-bottom: 1rem;
    }

    .nav-menu {
        gap: 1rem;
    }
}
```

```

    .page h1 {
      font-size: 2rem;
    }

    .stats-grid {
      grid-template-columns: 1fr;
    }
  }
}

```

4. components/Home.js

jsx

```

import React, { useContext } from 'react';
import { Link } from 'react-router-dom';
import { AuthContext } from '../App';

function Home() {
  const { user } = useContext(AuthContext);

  return (
    <div className="page">
      <h1>Welcome to Auth Flow Demo</h1>
      <p>
        This application demonstrates a complete authentication flow using React Router v6.
        {user ? ` You are currently logged in as ${user.username}.` : ' You are currently browsing as a guest.'}
      </p>

      <div className="features-grid">
        <div className="feature-card">
          <div className="feature-icon"> </div>
          <h3>Secure Login</h3>
          <p>Simulated authentication with protected routes</p>
        </div>

        <div className="feature-card">
          <div className="feature-icon"> </div>
          <h3>Programmatic Navigation</h3>
          <p>Using useNavigate hook for redirects</p>
        </div>

        <div className="feature-card">
          <div className="feature-icon">🔒</div>
          <h3>Route Protection</h3>
          <p>Conditional navigation with Navigate component</p>
        </div>

        <div className="feature-card">
          <div className="feature-icon"> </div>
          <h3>User Dashboard</h3>
          <p>Personalized content for authenticated users</p>
        </div>
      </div>

      <div className="cta-section">
        {!user ? (
          <>
            <h3>Ready to explore the dashboard?</h3>
            <p>Login to access your personalized dashboard with user-specific data.</p>
            <Link to="/login" className="cta-btn">Go to Login</Link>
          </>
        ) : (
          <>
            <h3>Welcome back, {user.username}!</h3>
            <p>Visit your dashboard to view your stats and activities.</p>
            <Link to="/dashboard" className="cta-btn">Go to Dashboard</Link>
          </>
        )}
      </div>
    </div>
  );
}

export default Home;

```

5. components/Login.js

jsx

```

import React, { useState, useContext } from 'react';
import { useNavigate, Link } from 'react-router-dom';
import { AuthContext } from '../App';

function Login() {
  const [formData, setFormData] = useState({
    username: '',
    password: ''
  });
  const [error, setError] = useState('');
  const [loading, setLoading] = useState(false);

  const { login } = useContext(AuthContext);
  const navigate = useNavigate();

  const handleChange = (e) => {
    setFormData({
      ...formData,
      [e.target.name]: e.target.value
    });
    // Clear error when user starts typing
    setError('');
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    setLoading(true);
    setError('');

    // Simulate API call delay
    setTimeout(() => {
      const { username, password } = formData;

      // Simple validation
      if (!username || !password) {
        setError('Please enter both username and password');
        setLoading(false);
        return;
      }

      // Attempt login
      const success = login(username, password);

      if (success) {
        // Programmatic navigation to dashboard
        navigate('/dashboard');
      } else {
        setError('Invalid credentials. Try any username/password.');
```

```

        disabled={loading}
      />
    </div>

    <div className="form-group">
      <label htmlFor="password">Password:</label>
      <input
        type="password"
        id="password"
        name="password"
        value={formData.password}
        onChange={handleChange}
        placeholder="Enter any password"
        disabled={loading}
      />
    </div>

    <button
      type="submit"
      className="login-btn"
      disabled={loading}
    >
      {loading ? 'Logging in...' : 'Login'}
    </button>

    <div className="demo-credentials">
      <p><strong>Demo Credentials:</strong></p>
      <p>Any username and password will work</p>
      <p>(Just don't leave them empty)</p>
    </div>
  </form>

  <p style={{ textAlign: 'center', marginTop: '1rem' }}>
    Don't have an account? <Link to="/" style={{ color: '#667eea' }}>Go to Home</Link>
  </p>
</div>
</div>
);
}

```

```
export default Login;
```

6. components/Dashboard.js

```

jsx

import React, { useContext } from 'react';
import { Navigate, Link } from 'react-router-dom';
import { AuthContext } from '../App';

function Dashboard() {
  const { user, logout } = useContext(AuthContext);

  // Conditional navigation - if not authenticated, redirect to login
  if (!user) {
    return <Navigate to="/login" replace />;
  }

  // Mock data for dashboard
  const stats = {
    posts: 24,
    followers: 1250,
    following: 342,
    likes: 5432
  };

  const recentActivities = [
    'Updated profile picture',
    'Posted a new article',
    'Liked 5 posts',
    'Followed 3 new users',
    'Commented on a discussion'
  ];

  return (
    <div className="page">
      <div className="dashboard-header">
        <div>
          <h1>Dashboard</h1>
          <div className="user-info">

```



```

        Welcome back, {user.username}! ({user.email})
    </div>
</div>
<button onClick={logout} className="logout-btn">
    Logout
</button>
</div>

<div className="stats-grid">
    <div className="stat-card">
        <h3>Total Posts</h3>
        <div className="stat-value">{stats.posts}</div>
    </div>

    <div className="stat-card">
        <h3>Followers</h3>
        <div className="stat-value">{stats.followers}</div>
    </div>

    <div className="stat-card">
        <h3>Following</h3>
        <div className="stat-value">{stats.following}</div>
    </div>

    <div className="stat-card">
        <h3>Total Likes</h3>
        <div className="stat-value">{stats.likes}</div>
    </div>
</div>

<div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '1.5rem' }}>
    <div className="activity-list">
        <h3>Recent Activities</h3>
        <ul>
            <li>
                {recentActivities.map((activity, index) => (
                    <li key={index}>{activity}</li>
                ))}
            </li>
        </ul>
    </div>

    <div style={{ background: '#f8f9fa', padding: '1.5rem', borderRadius: '8px' }}>
        <h3 style={{ marginBottom: '1rem' }}>Quick Actions</h3>
        <div style={{ display: 'flex', flexDirection: 'column', gap: '1rem' }}>
            <button style={{
                padding: '0.75rem',
                background: '#667eea',
                color: 'white',
                border: 'none',
                borderRadius: '4px',
                cursor: 'pointer'
            }}>
                Create New Post
            </button>
            <button style={{
                padding: '0.75rem',
                background: '#28a745',
                color: 'white',
                border: 'none',
                borderRadius: '4px',
                cursor: 'pointer'
            }}>
                View Profile
            </button>
            <button style={{
                padding: '0.75rem',
                background: '#17a2b8',
                color: 'white',
                border: 'none',
                borderRadius: '4px',
                cursor: 'pointer'
            }}>
                Settings
            </button>
        </div>
    </div>
</div>

<p style={{ marginTop: '2rem', textAlign: 'center', color: '#666' }}>
    <Link to="/" style={{ color: '#667eea' }}>< Back to Home</Link>
</p>
</div>

```

```
    );  
  }  
  
  export default Dashboard;  
}
```

7. index.js

```
import React from 'react';  
import ReactDOM from 'react-dom/client';  
import App from './App';  
  
const root = ReactDOM.createRoot(document.getElementById('root'));  
root.render(  
  <React.StrictMode>  
    <App />  
  </React.StrictMode>  
);
```

Output Screenshots (Text Description)

Home Page (Unauthenticated):

text

NAVBAR

Auth Flow Demo | Home | Dashboard |

WELCOME TO AUTH FLOW DEMO

=====

This application demonstrates a complete authentication flow using React Router v6.
You are currently browsing as a guest.

Secure Login
Simulated
authentication

Programmatic
Navigation
useNavigate hook

Route Protection
Conditional
Navigate

User Dashboard
Personalized
content

READY TO EXPLORE THE DASHBOARD?

=====

Login to access your personalized dashboard with user-specific data.

[Go to Login]

Login Page:

text

NAVBAR

Auth Flow Demo | Home | Dashboard |

LOGIN

=====

 Demo Credentials:
Any username and password will work
(Just don't leave them empty)

Username: [_____]

Password: [_____]

[Login]

Don't have an account? Go to Home

Dashboard Page (Authenticated):

text

NAVBAR

Auth Flow Demo | Home | Dashboard |

DASHBOARD

=====

Welcome back, john_doe! (john_doe@example.com) [Logout]

Total Posts24

Followers1250

Following342

Total Likes5432

RECENT ACTIVITIES

- Updated profile
- Posted article
- Liked 5 posts
- Followed 3 users
- Commented on post

QUICK ACTIONS

[Create New Post]

[View Profile]

[Settings]

← Back to Home

After Logout (Redirect to Login):

text

NAVBAR

Auth Flow Demo | Home | Dashboard |

(You are automatically redirected to login page)

Key Features Implemented:

1. Authentication Context: Using React Context API for global auth state
2. Programmatic Navigation: useNavigate hook for redirect after login
3. Route Protection: Navigate component for conditional redirects
4. Login Simulation: Mock authentication with any credentials
5. Protected Dashboard: Only accessible when authenticated
6. Logout Functionality: Clears user state and redirects to login
7. Responsive Design: Mobile-friendly layout with CSS grid/flexbox

Authentication Flow:

1. Unauthenticated User:

Can access Home and Login pages

Dashboard attempts redirect to Login
2. Login Process:

User enters any credentials

On submit, useNavigate redirects to Dashboard

- Auth context updates with user info

3. Authenticated User:

- Can access Dashboard
- Dashboard shows user-specific data
- Logout button clears session

4. Route Protection:

- Dashboard checks auth status
- If no user, `<Navigate to="/login" />` redirects

How to Run:

1. Create React app: `npx create-react-app react-auth-flow`
2. Install React Router: `npm install react-router-dom`
3. Replace files with code above
4. Run: `npm start`
5. Visit <http://localhost:3000>

The application demonstrates a complete authentication flow with protected routes, programmatic navigation, and conditional rendering based on authentication status.